GATE ME 90 Days Challenge

Day 36/90 · Practice Set with Detailed Solutions

SUBJECT	TODAY'S TOPIC	COUNTDOWN
Weekly Revision	Week 6 Revision	70 days to GATE

How to use this set: 15 questions (6 MCQ, 9 NAT) on all five topics of Week 6: momentum, flow measurement and dimensional analysis; viscous flow and boundary layer; analytical reasoning; turbulent flow and pipe losses; thermodynamics basics and pure substances (3 questions each). Try them in **45 minutes** without looking at the solutions, then check the answer key and the step-by-step solutions. For NAT questions, enter the number; there is no negative marking for NAT in GATE.
Today's test: open 7 to 9 PM in the **GATE ME 90 Days Challenge** course on our app and website (enrol once for ₹1: iurlk.courses.store/909124), 20 new questions on the same topic. Top scorers are announced in the next 8 AM challenge reel.

Questions

Q1 NAT

A pitot-static tube is placed at the centre of an air duct. The differential water manometer connected to it reads 40 mm. Taking the density of air as 1.2 kg/m³, the density of water as 1000 kg/m³, the pitot-tube coefficient as 0.98 and g = 9.81 m/s², the air velocity (in m/s, round off to two decimal places) is _____

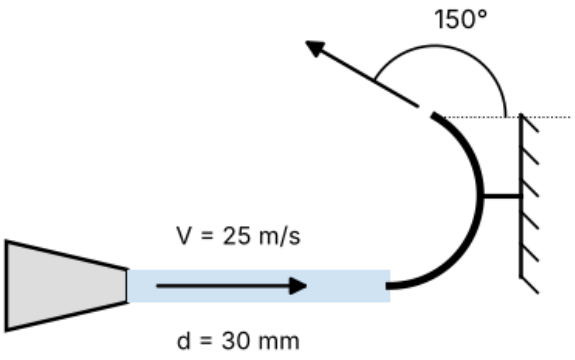
Q2 MCQ

The drag on a submarine moving at 2 m/s in sea water is to be found from a 1 : 20 model tested in the same sea water under Reynolds similarity. The ratio of the drag on the model to the drag on the prototype is

(A) 1/400
(B) 1/20
(C) 1
(D) 20

Q3 NAT

A 30 mm diameter water jet with velocity 25 m/s strikes a fixed, smooth curved vane and is deflected through 150°, as shown. Taking the density of water as 1000 kg/m³ and neglecting friction, the force exerted by the jet on the vane in the direction of the jet (in N, round off to one decimal place) is _____



Q4 NAT

A shaft of 80 mm diameter rotates at 1200 rpm in a journal bearing 120 mm long. The radial clearance of 0.1 mm is filled with oil of dynamic viscosity 0.08 Pa·s. Assuming a linear velocity profile in the oil film, the power lost in the bearing (in W, round off to the nearest integer) is _____

Q5 MCQ

For fully developed laminar flow in a circular pipe, the kinetic-energy correction factor α and the momentum correction factor β are, respectively,

(A) 1.0 and 1.0
(B) 2.0 and 1.33
(C) 1.5 and 1.2
(D) 1.33 and 2.0

Q6 NAT

A thin flat plate 1.5 m long (in the flow direction) and 0.6 m wide is towed edgewise at 0.2 m/s through still water (density 1000 kg/m³, kinematic viscosity 1.0×10^{-6} m²/s). Using the Blasius result $C_D = 1.328/\sqrt{Re_L}$ for a laminar boundary layer, the total skin-friction drag on **both sides** of the plate (in N, round off to three decimal places) is _____

Q7 MCQ

In a certain code, FLOW is written as GNRA. How is PUMP written in that code?

(A) QWPT
(B) QVNQ
(C) QWPS
(D) RWPT

Q8 MCQ

Pointing to a girl in a photograph, Ravi says, "She is the daughter of the only son of my grandfather." Assuming Ravi's father is that only son, how is the girl related to Ravi?

(A) Cousin
(B) Niece
(C) Sister
(D) Aunt

Q9 NAT

In a class of 60 students, 35 study Thermodynamics, 30 study Fluid Mechanics and 12 study neither. The number of students who study both subjects is _____

Q10 NAT

A pipeline consists of 600 m of 300 mm diameter pipe followed by 400 m of 200 mm diameter pipe, both with the same friction factor. Neglecting minor losses, the length of a single 250 mm diameter pipe (same friction factor) that gives the same head loss at the same discharge (in m, round off to the nearest integer) is _____

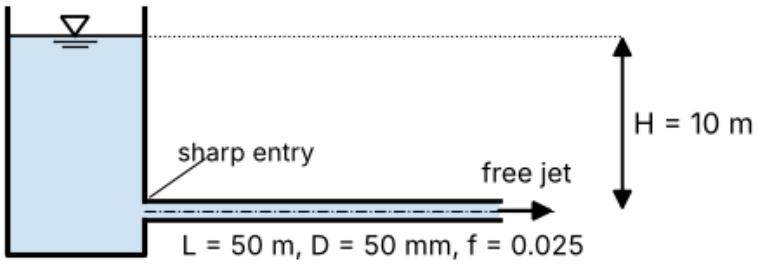
Q11 MCQ

In turbulent pipe flow the thickness of the viscous sublayer is 0.05 mm, and the average height of the wall roughness is 0.15 mm. According to the usual criterion (smooth if $\varepsilon/\delta' < 0.25$, rough if $\varepsilon/\delta' > 6$), the pipe behaves as

(A) hydraulically smooth
(B) hydraulically rough
(C) in the transition regime between smooth and rough
(D) laminar

Q12 NAT

Water flows from a large open tank through a 50 m long, 50 mm diameter horizontal pipe (Darcy f = 0.025) with a sharp-edged entrance (K = 0.5) and discharges as a free jet into the atmosphere, as shown. The water level is 10 m above the pipe centreline. Taking g = 9.81 m/s², the discharge (in litres per second, round off to two decimal places) is _____



Q13 MCQ

On a new temperature scale °N, the ice point is 25 °N and the steam point is 175 °N, with the scale linear in Celsius. The temperature at which the Celsius and °N readings are numerically equal is

(A) −50
(B) 50
(C) −40
(D) −25

Q14 NAT

A gas in a piston–cylinder device, restrained by a linear spring, expands so that its pressure varies linearly with volume from 150 kPa at 0.02 m³ to 450 kPa at 0.06 m³. The work done by the gas (in kJ) is _____

Q15 NAT

Superheated steam at 10 MPa and 400 °C has a specific volume of 0.02641 m³/kg (from steam tables). Taking the gas constant of steam as 0.4615 kJ/kg·K, the compressibility factor Z of the steam (round off to three decimal places) is _____

Answer Key

Q	Type	Answer
Q1	NAT	25.06 (accept 24.81 to 25.31)
Q2	MCQ	(C) 1
Q3	NAT	824.4 (accept 820.3 to 828.5)
Q4	NAT	610 (accept 604 to 616)
Q5	MCQ	(B) 2.0 and 1.33
Q6	NAT	0.087 (accept 0.086 to 0.088)
Q7	MCQ	(A) QWPT
Q8	MCQ	(C) Sister
Q9	NAT	17
Q10	NAT	1462 (accept 1447 to 1477)
Q11	MCQ	(C) in the transition regime between smooth and rough
Q12	NAT	5.34 (accept 5.29 to 5.40)
Q13	MCQ	(A) −50
Q14	NAT	12
Q15	NAT	0.850 (accept 0.846 to 0.855)

Detailed Solutions

Q1 **Answer: 25.06 (accept 24.81 to 25.31)**

Concept: A pitot-static tube measures the dynamic pressure $\Delta p = \tfrac{1}{2}\rho_{air}V^2$, and $V = C\sqrt{(2\Delta p/\rho_{air})}$.

$\Delta p = \rho_w g h = 1000 \times 9.81 \times 0.040 = 392.4$ Pa (the air column in the manometer is negligible)

$V = 0.98 \times \sqrt{(2 \times 392.4/1.2)} = 0.98 \times 25.57 =$ **25.06 m/s**

Exam tip: Use the density of the **flowing** fluid (air) under the square root, and the manometer liquid only to convert the reading into Δp.

Q2 **Answer: (C) 1**

Concept: A fully submerged body has no free-surface waves, so viscous forces dominate and Reynolds similarity applies.

Same fluid: $V_m L_m = V_p L_p \rightarrow V_m = 20 \times 2 = 40$ m/s

$F \propto \rho V^2 L^2$, so $F_m/F_p = (V_m/V_p)^2 (L_m/L_p)^2 = 20^2 \times (1/20)^2 =$ **1**

Exam tip: Under Reynolds similarity with the same fluid, forces are equal in model and prototype; that is why such model tests need very high speeds or a different fluid.

Q3 Answer: 824.4 (accept 820.3 to 828.5)

Concept: Momentum equation along the jet axis: $F = \dot{m}(V_{in,x} - V_{out,x})$. On a smooth fixed vane the speed stays V.

$\dot{m} = \rho AV = 1000 \times (\pi/4)(0.03)^2 \times 25 = 17.67$ kg/s

$V_{out,x} = V\cos150° = -0.866V$, so $F = \dot{m}V(1 + \cos30°) = 17.67 \times 25 \times 1.866 =$ **824.4 N**

Exam tip: A 180° turn (U-shaped cup, as in a Pelton bucket) gives the maximum, $2\rho AV^2$; a flat plate normal to the jet gives ρAV^2.

Q4 Answer: 610 (accept 604 to 616)

Concept: Thin oil film → Couette flow: $\tau = \mu U/h$. Viscous torque = $\tau \times (\pi DL) \times D/2$, power = $T\omega$.

$U = \pi DN/60 = \pi \times 0.08 \times 1200/60 = 5.027$ m/s; $\tau = 0.08 \times 5.027/0.0001 = 4021$ Pa

$F = \tau\pi DL = 4021 \times \pi \times 0.08 \times 0.12 = 121.3$ N; $T = F \times 0.04 = 4.851$ N·m

$P = T\omega = 4.851 \times 125.66 =$ **610 W**

Exam tip: $P \propto \mu N^2D^3L/h$: doubling the speed quadruples the loss, halving the clearance doubles it.

Q5 Answer: (B) 2.0 and 1.33

Concept: $\alpha = (1/A)\int(u/V)^3dA$ and $\beta = (1/A)\int(u/V)^2dA$, with the parabolic profile $u = 2V(1 - r^2/R^2)$.

Integrating gives $\alpha =$ **2** and $\beta =$ **4/3 ≈ 1.33**.

Turbulent profiles are much flatter: $\alpha \approx 1.03$–1.1 and $\beta \approx 1.01$–1.04.

Exam tip: Always $\alpha > \beta > 1$ for a non-uniform profile; both equal 1 only for uniform flow.

Q6 Answer: 0.087 (accept 0.086 to 0.088)

Concept: Check that the boundary layer is laminar ($Re_L < 5 \times 10^5$), then $F = C_D \times \frac{1}{2}\rho U^2 \times$ area.

$Re_L = UL/\nu = 0.2 \times 1.5/10^{-6} = 3 \times 10^5$ (laminar over the whole plate)

$C_D = 1.328/\sqrt{(3 \times 10^5)} = 0.002425$

$F = 0.002425 \times \frac{1}{2} \times 1000 \times 0.2^2 \times (1.5 \times 0.6) \times 2 =$ **0.087 N**

Exam tip: Don't forget the factor 2 for both faces; the local coefficient at x is $0.664/\sqrt{Re_x}$, exactly half the average coefficient.

Q7 Answer: (A) QWPT

Concept: Compare letter positions: F→G (+1), L→N (+2), O→R (+3), W→A (+4, wrapping past Z).

Apply the same +1, +2, +3, +4 to PUMP: P→Q, U→W, M→P, P→T

Code = **QWPT**

Exam tip: Write the alphabet positions (A = 1 … Z = 26) once and count forward; remember to wrap Z → A.

Q8 Answer: (C) Sister

Concept: Decode the chain from the far end.

The only son of Ravi's grandfather = Ravi's father.

The daughter of Ravi's father = Ravi's **sister**.

Exam tip: 'Only son' removes uncles from the picture; draw a small family tree with + and − for male and female.

Q9 **Answer: 17**

Concept: $n(A \cup B) = n(A) + n(B) - n(A \cap B)$.

Students studying at least one subject = 60 − 12 = 48

48 = 35 + 30 − n(both) → n(both) = **17**

Exam tip: Check: only Thermo = 18, only Fluids = 13, both = 17, neither = 12; total 60.

Q10 **Answer: 1462 (accept 1447 to 1477)**

Concept: At fixed Q and f, $h_f = 8fLQ^2/(\pi^2 g D^5) \propto L/D^5$ (Dupuit's equation for equivalent pipes).

$L_e/D^5 = L_1/D_1^5 + L_2/D_2^5 \rightarrow L_e = 600(250/300)^5 + 400(250/200)^5$

$L_e = 600 \times 0.4019 + 400 \times 3.0518 = 241.1 + 1220.7 =$ **1462 m**

Exam tip: The small-diameter segment dominates: halving D raises the loss 32 times.

Q11 **Answer: (C) in the transition regime between smooth and rough**

Concept: Compare the roughness height ε with the viscous-sublayer thickness $\delta' = 11.6\nu/u^*$.

$\varepsilon/\delta' = 0.15/0.05 = 3$, which lies between 0.25 and 6.

So the pipe is in the **transition regime**: f depends on both Re and ε/D (Colebrook).

Exam tip: The same pipe can be smooth at low Re and rough at high Re, because δ′ shrinks as velocity rises.

Q12 **Answer: 5.34 (accept 5.29 to 5.40)**

Concept: Energy equation from the free surface to the jet: $H = V^2/2g$ (jet) + $0.5V^2/2g$ (entry) + $fLV^2/(2gD)$ (friction).

$10 = (V^2/2g)(1 + 0.5 + 0.025 \times 50/0.05) = (V^2/2g)(26.5)$

$V = \sqrt{(2 \times 9.81 \times 10/26.5)} = 2.721$ m/s

$Q = (\pi/4)(0.05)^2 \times 2.721 = 5.343 \times 10^{-3}$ m³/s = **5.34 L/s**

Exam tip: For a free jet, the 1.0 in the bracket is the jet's kinetic energy; for a pipe ending under water, it is the exit loss. The total is the same.

Q13 **Answer: (A) −50**

Concept: Two linear scales are related by (N − ice)/(steam − ice) on each scale.

(N − 25)/(175 − 25) = C/100 → N = 25 + 1.5C

Equal readings: t = 25 + 1.5t → t = **−50**

Exam tip: The zeroth law makes such scales possible: any thermometric property gives a linear scale once two fixed points are chosen. For °C and °F the equal reading is −40.

Q14 **Answer: 12**

Concept: Boundary work $W = \int p\,dV$ is the area under the process line on the p–V diagram.

For a straight line the area is a trapezium: $W = \frac{1}{2}(p_1 + p_2)(V_2 - V_1)$

$W = \frac{1}{2} \times (150 + 450) \times (0.06 - 0.02) =$ **12 kJ**

Exam tip: kPa × m³ = kJ. Work and heat are path functions: the same end states joined by a different path give different work.

Q15 **Answer: 0.850 (accept 0.846 to 0.855)**

Concept: Z = pv/(RT) measures the departure from ideal-gas behaviour (Z = 1 for an ideal gas).

Z = (10 000 × 0.02641)/(0.4615 × 673.15) = 264.1/310.66 = **0.850**

So the ideal-gas equation would overpredict the specific volume by about 1/0.85 − 1 ≈ 18 %.

Exam tip: Steam behaves as an ideal gas only at low pressure (roughly below 10 kPa) or very high temperature; near the saturation dome use the tables.

What next?

Day 36 test tonight, 7 to 9 PM

20 questions · 60 minutes · GATE marking (1/3 and 2/3 negative). Top 3 scorers are announced with name, marks and photo (if shared) in the next 8 AM challenge reel. Not enrolled yet? Join the ₹1 challenge course: iurlk.courses.store/909124

Website test series

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Next, Day 37/90: Thermodynamics, First Law. Comment **GATE90** on our reel to get all links.
Want concise revision notes? The GATE ME handwritten short notes of an AIR 16 topper are available in the Dig Career Thrust app.